PHYSICAL CUME
April 26, 2002

1. Give a sketch and the molecular point group for the following molecules. Also indicate if the molecule may have a dipole moment.

   (a) Nitrous oxide N_2O (note: the O atom is not in the middle!)
   (b) $Ni(CO)_4$ (non-planar)
   (c) Trans-butadiene
   (d) Chair cyclohexane
   (e) Sulfur trioxide (planar)
   (f) Sulfur dioxide (non-linear!)
   (g) 1,3,5-trifluorobenzene
   (h) methyl iodide
   (i) acetylene
   (j) $Fe(CN)_6^{4-}$ (the hexacyanoferrate(II) anion)

2. The character table of the group C_{2v} is given below.

C_{2v}	E	C_2	$\sigma(xz)$	$\sigma(yz)$		
A_1	1	1	1	1	T_z	α_{xx}, α_{yy}, α_{zz}
A_2	1	1	-1	-1	R_z	α_{xy}
B_1	1	-1	1	-1	T_x; R_y	α_{xz}
B_2	1	-1	-1	1	T_y; R_x	α_{yz}

(a) Describe the symmetry operations and irreducible representations (symmetry species) for this group.

(b) The ground state of the molecule belongs to the A_1 representation. The excited states can be of any of the four irreducible representations. Which of the following transitions originating at the ground state are dipole-allowed?

$A_1 \leftrightarrow A_1$ $\quad$ $A_1 \leftrightarrow A_2$ $\quad$ $A_1 \leftrightarrow B_1$ $\quad$ $A_1 \leftrightarrow B_2$